Probability and Random Variables

Expectations and Variances of Standard Random Variables

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Uniform Random Variables

- $P(X = k) = \frac{1}{b-a+1}$ for $k = a, \cdots, b$
- $E[X] = \frac{a+b}{2}$

- $\mathrm{var}(X) = \frac{(b-a+1)^2-1}{12}$

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Bernoulli Random Variables

- $P(X = k) = \begin{cases} 1 - p, & \text{if } k = 0 \\ p, & \text{if } k = 1 \end{cases}$
- $E[X] = p$
- $\text{var}(X) = p(1 - p)$

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Binomial Random Variables

- $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ for $k = 0, \cdots, n$
- $E[X] = np$

- $\mathrm{var}(X) = np(1-p)$

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Geometric Random Variables

- $P(X = k) = (1-p)^{k-1} \cdot p$ for $k = 1,2,3, \cdots$
- $E[X] = \frac{1}{p}$

- $\mathrm{var}(X) = \frac{1-p}{p^2}$

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Poisson Random Variables

- $P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$ for $k = 0,1,2,\cdots$
- $E[X] = \lambda$

- $\mathrm{var}(X) = \lambda$

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